LESSON 1.5

Math is Finding Patterns

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

WHY THIS MATTERS Akela's change jar is full of pennies, nickels, dimes, and quarters. The value of the coins is $53.50. How many of each coin could be in the jar? There is more than one answer — and patterns are the way in.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
pattern *Spanish: patrón*	Something that repeats or changes in a predictable way.	1, 3, 7, 15, 31 — each tower's steps follow a predictable jump.
pattern rule *Spanish: regla del patrón*	The instruction that tells how to get from one number in a pattern to the next.	2, 4, 6, 8 … the rule is "+ 2 every time".
generalization *Spanish: generalización*	A statement that is true for a whole pattern, made by noticing repeated calculations.	1 → 3, 2 → 5, 3 → 7 … so n → 2n + 1, for ANY n.
table of values *Spanish: tabla de valores*	A table that lists each step of a pattern so you can check that a solution is reasonable.	discs 1, 2, 3 beside steps 1, 3, 7 — one row per step.
reasonableness *Spanish: sensatez del resultado*	Whether an answer makes sense — checked as you work, with adjustments made when needed.	5/6 of 540 should land BELOW 540, so 650 fails the check.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A change jar is full of pennies, nickels, dimes, and quarters worth $53.50 in total. How many of each coin could there be?

1. Each penny has a value of $0.01, so 100 pennies have a value of $1.00.
2. Each nickel has a value of $0.05, so 100 nickels have a value of $5.00. Each dime has a value of $0.10, so 100 dimes have a value of $10. Each quarter has a value of $0.25, so 100 quarters have a value of $25.
3. I multiply the number of each coin by its value. And I notice a generalization: because the jar's total, $53.50, ends in 0, the number of pennies can only be a multiple of 5.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Try 200 quarters: 200 × $0.25 = $50.00.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- _____ is only a one-time fact because _____; it would be a pattern if _____.
- I noticed that _____ .
- This helps because instead of counting one at a time, I can _____ .
- I look for _____ .

Word bank pattern | pattern rule | generalization | table of values | reasonableness

WATCH OUT!

A common mistake

Reporting only the points from the streak (20 or 30) and forgetting to add the points each player had before the streak began.

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Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Each penny has a value of $0.01. What is the value of 100 pennies?

(A) $0.10
(B) $1.00
(C) $10.00
(D) $100.00

HINT A penny is one hundredth of a dollar.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Each quarter has a value of $0.25. Following the pattern, what is the value of 100 quarters?

(A) $2.50
(B) $25.00
(C) $100.25
(D) $250.00

HINT Multiplying by 100 moves the decimal point.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Player A scores a two-pointer on every shot. What is Player A's pattern rule?

(A) Add 2
(B) Multiply by 2
(C) Add 3
(D) Add 18

HINT How much is one two-pointer worth?

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Player B had 6 points, then made 10 three-pointers in a row. How many points does Player B have now?

(A) 9
(B) 16
(C) 30
(D) 36

HINT How many points do 10 three-pointers add?

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Name a pattern you noticed in the coin values, and how it could help you count money faster.

SENTENCE STARTER I noticed that ___ .
This helps because instead of counting one at a time, I can ___ .

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

Where do you look for a pattern first when a problem feels too big to count?

SENTENCE STARTER I look for ___ .
A repeated calculation helps because ___ .

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a card you called 'A one-time fact.' What would have to keep happening for it to become a generalization instead — and how would you write that as a pattern rule?

SENTENCE STARTER ___ is only a one-time fact because ___; it would be a pattern if ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.5

Math is Finding Patterns

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find patterns and pattern rules, use them to make generalizations, and check my solutions with a table of values.

WHY THIS MATTERS Akela's change jar is full of pennies, nickels, dimes, and quarters. The value of the coins is $53.50. How many of each coin could be in the jar? There is more than one answer — and patterns are the way in.

PART 1 Apply it

real-world problems

1 **Circle the letter of the best answer. Show how you know.**

Player A had 18 points before making 10 two-pointers in a row. How many points does Player A have after the streak?

(A) 20
(B) 28
(C) 38
(D) 48

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Akela's jar totals $53.50. Why can the number of pennies only be a multiple of 5?

(A) Because the total's cents end in 0, and only pennies make amounts that aren't multiples of 5 cents
(B) Because pennies always come in rolls of 5
(C) Because there are exactly 5 kinds of coins in the jar
(D) Because $53.50 is divisible by 5

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which combination of coins totals exactly $53.50?

(A) 100 quarters, 25 dimes, 15 nickels, 25 pennies

(B) 200 quarters, 25 dimes, 15 nickels, 25 pennies

(C) 200 quarters, 25 dimes, 15 nickels, 20 pennies

(D) 200 quarters, 30 dimes, 15 nickels, 25 pennies

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How does a table of values check that Player A really scored 38 points?

(A) It rounds the answer to the nearest ten

(B) It replaces the pattern rule so you don't need one

(C) It proves the answer without doing any calculations

(D) It lists the score after every shot, so you can see the pattern land exactly on 38

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

5 **Answer in complete sentences.**

Use the multiple-of-5 pattern to find a DIFFERENT combination of coins that totals $53.50. Show that it works.

WRITE YOUR ANSWER

6 **Answer in complete sentences.**

After the streaks, Player A has 38 and Player B has 36. Player B started 12 points behind but nearly caught up. What about the two pattern rules explains that?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a card you called 'A one-time fact.' What would have to keep happening for it to become a generalization instead — and how would you write that as a pattern rule?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 1.5

Math is Finding Patterns

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each input-output table to its pattern rule.

_____ In: 1, 2, 3 → Out: 5, 10, 15

_____ In: 1, 2, 3 → Out: 3, 5, 7

_____ In: 2, 4, 6 → Out: 1, 2, 3

_____ In: 1, 2, 3 → Out: 2.5, 5, 7.5

(A) Multiply by 5

(B) Double, then add 1

(C) Divide by 2

(D) Multiply by 2.5

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

If both players kept their streaks going, after how many MORE shots would Player B tie Player A?

(A) 1 more shot

(B) They can never tie

(C) 2 more shots

(D) 10 more shots

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

A sei whale measures 4.5 meters at birth and grows about 2.5 centimeters each day. About how many days until it reaches 15 meters?

(A) About 4.2 days
(B) About 42 days
(C) About 420 days
(D) About 600 days

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

4 **Answer in complete sentences.**

Write a generalization about ANY player who makes n shots in a row worth p points each, starting from s points. Then test it on Player A.

WRITE YOUR ANSWER

5 **Answer in complete sentences.**

The lesson says we 'evaluate the reasonableness of solutions as we work and make adjustments.' Describe a moment solving the coin-jar problem where you would stop and adjust.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Take a card you called 'A one-time fact.' What would have to keep happening for it to become a generalization instead — and how would you write that as a pattern rule?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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